

E#5 - Module 5: In-Class Exercises #5

CSC6033 – Week 5

In-Class Exercise E#5

CFG and PDA for Palindromes (Even and Odd Length)

1. Introduction

This document presents a **Context-Free Grammar (CFG)** and a **Pushdown Automaton (PDA)** that accept palindromes of **both even and odd lengths** over the alphabet $\Sigma = \{a, b\}$. The solution merges the concepts from **Examples 3 and 4** (slides 8, 9, 14, 15, 28, 29).

2. Context-Free Grammar (CFG)

Alphabet and Components

- **Terminals (Σ):** $\{a, b\}$
- **Non-terminals (V):** $\{S\}$
- **Start symbol:** S

Productions

$S \rightarrow aSa \mid bSb \mid a \mid b \mid \varepsilon$

Explanation

- aSa and bSb ensure symmetry for palindromes.
- a and b allow a single middle character (odd length).
- ε allows empty string (even length).

3. Pushdown Automaton (PDA)

Formal Definition Formal PDA 7-Tuple Summary:

$M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$ where:

$Q = \{I, J, K, F\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, b, z\}$

δ = transition function (see table)

$q_0 = I$

z = initial stack symbol

$F = \{F\}$

- **States:** $\{I, J, K, F\}$
- **Input symbols:** $\{a, b\}$
- **Stack symbols:** $\{a, b, z\}$ (z = bottom marker)
- **Start state:** I
- **Accepting state:** F

Transition Table

Current State	Input	Stack Top	Action	Next State
I	ϵ	ϵ	push z	J
J	a	ϵ	push a	J
J	b	ϵ	push b	J
J	a	a	pop	J
J	b	b	pop	J
J	a	z	skip middle letter	K
J	b	z	skip middle letter	K
J	ϵ	z	pop z	F
K	a	a	pop	K
K	b	b	pop	K

Current State	Input	Stack Top	Action	Next State
K	ϵ	z	pop z	F

How It Works

- Start by pushing z as a bottom marker.
- For each input symbol:
 - Non-deterministically **push** (first half) or **pop** (second half).
- For odd palindromes, allow skipping one unmatched letter (middle).
- Accept if stack returns to z and input is consumed.

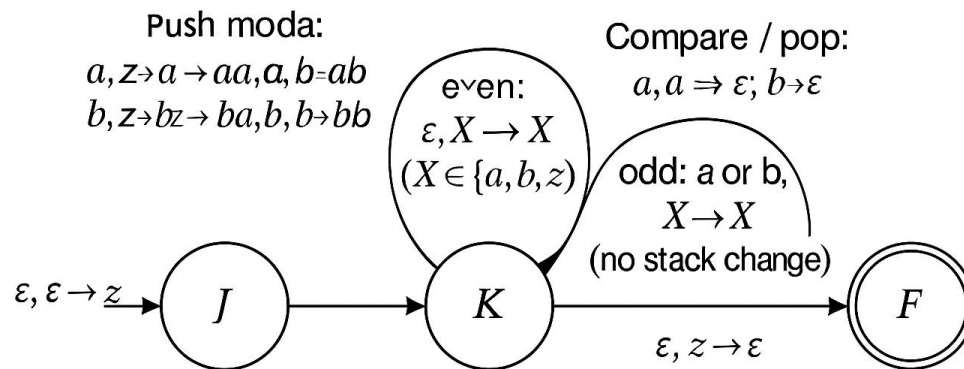
4. PDA Diagram (Text Description)

- **State I:** Push z and go to J.
- **State J:** Read letters, push or pop based on guess.
- **Optional transition:** Skip one letter for an odd palindrome.
- **State K:** Continue popping until z.
- **State F:** Accept when the stack has only z.

5. Remarks on Adaptation

- Combined rules from Example 3 (even palindromes) and Example 4 (odd palindromes).
- CFG includes ϵ for even and single letters for odd.
- PDA is **non-deterministic** because it guesses when to switch from pushing to popping.
- This PDA accepts both even and odd palindromes over {a, b}.

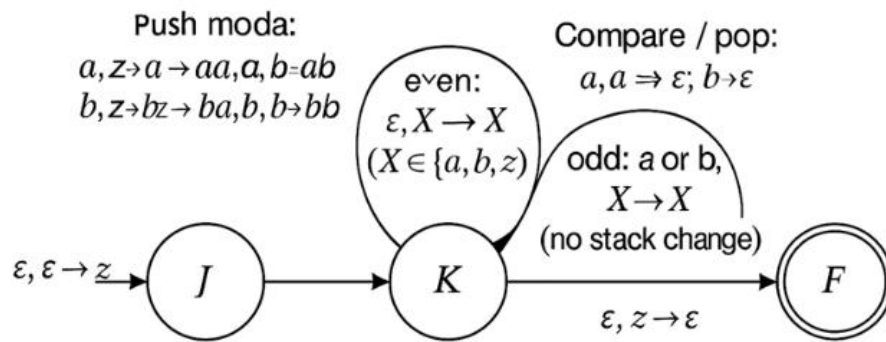
3.2 PDA diagram



NPDA for palindromes over $\{a,b\}$
(even and odd lengths)

3.2 PDA Diagram

Below is a visual representation of the PDA:



NPDA for palindromes over $\{a, b\}$
 (even and odd lengths)

Figure: Visual diagram of NPDA for palindromes over $\{a, b\}$ (even and odd lengths)

6. Example Runs

- abba $\rightarrow$ accepted (even palindrome)
- aba $\rightarrow$ accepted (odd palindrome)
- abab $\rightarrow$ rejected (not a palindrome)

7. Quick Acceptance Tests

Accepted: ε , a, b, aa, bb, aba, abba

Rejected: ab, abb, abab

8. Conclusion

This exercise demonstrates how CFGs and PDAs can be combined to recognize palindromes of varying lengths. The use of nondeterminism allows flexible midpoint handling, making the PDA capable of accepting both even and odd palindromes. The visual diagram and example runs help validate the correctness of the design.